Milestone Checklist

Please note that documentation and code reviews are a huge portion of the life cycle of a model.
Please refer here for documentation timeline checklist: Document Milestone Checklist.

Milestone POCs

Role	POC
Architecture Software Lead	Reid, Jessica (Durandal) Havran, Jacob [USA]
Architecture Domain Lead	Perrin, Josiah (MBO Consultant)
SERAPH Team Lead	Schaeffer, Tiago [USA]
TDL Tech Lead	Keene, April (MBO)
Tech Editor	Childers, Joe [USA]

System TDL Planning

- [ ] TDL Tech lead to identify system model lead and team composition with technical lead input
- [ ] Identify system stakeholders (NGIC Analyst, NGIC Product Owner)
- [ ] Identify when the model is due (FY24, etc.)
- [ ] Verify intel has been sent to Booz Allen

Pre-Kickoff

- [ ] Model lead to do deep review on available intel
  - [ ] Model lead to identify important intel gaps
- [ ] Model lead to identify any additional SME required
- [ ] Develop questions around the system of interest
- [ ] Antenna Review (develop questions for kickoff)
- [ ] Develop a list of system capabilities and questions
  - [ ] Review UML with Architecture Software Lead to highlight any potential architecture gaps ahead of kickoff meeting
- [ ] Understand the system-of-systems impact of the system of focus where possible
- [ ] Identify template model to use
- [ ] Determine required attendees for the kickoff meeting and provide to NGIC
  - [ ] Reserve Booz Allen conference room
- [ ] Model lead gets concurrence from TDL tech lead that the team is ready for kickoff

Kickoff: Virtual or In-person

- [ ] Gather additional intelligence documents
- [ ] Gather input from analyst
- [ ] Initial Scoping of requirements
- [ ] Initial cut on requirements spreadsheet (NOTE: This is kept internal only as a reference to make sure all questions and assumptions discussed)
- [ ] Plan for the unclassified to classified discussion
- [ ] Identify any systems associated with this threat system and their model development timeline
- [ ] Identify any similar systems to the threat of interest and do any models exist of those systems

- [ ] Determine if integration into another model is a requirement
  - [ ] ITASE
  - [ ] AFSIM
  - [ ] C4
  - [ ] Other intel models
- [ ] Architecture Description Document (ADD)
  - [ ] Drafts:
    - [ ] Introduction
    - [ ] Architecture Selection Criteria
    - [ ] Architecture Version
    - [ ] Architecture Standards and Regulations
    - [ ] Model Standards and Regulations
    - [ ] Supported Software and Operating Systems
    - [ ] Threat Model Class Diagram
- [ ] User and Analyst Manual (UAM)
  - [ ] Drafts:
    - [ ] Introduction
    - [ ] Model Functional Block Diagram

Post-Kickoff

- [ ] Develop a bi-weekly (every other week) technical meeting with model stakeholders
- [ ] If integration is required, determine who the model lead is for that work
- [ ] Develop a meeting schedule with architecture integration leads (ITASE, AFSIM, other intel models)
- [ ] Requirements Spreadsheet: Update Initial Draft and send to NGIC
  - [ ] May require back and forth with NGIC to get to 99%
- [ ] Build out Epics for unclassified and epics for classified based on mostly finalized req sheet
- [ ] Identify any gaps in the model architecture and work with architecture lead
- [ ] Once requirements spreadsheet in good shape, build out ROM for the model cost

SRR: Virtual or In-person

- [ ] Refer to document templates for expectations
- [ ] Requirements Spreadsheet: Initial Draft - Expected to be 99% complete prior to official meeting with minor changes during meeting **(this is expected to be classified)**
- [ ] Programmatics performed by PM team
  - [ ] Project Plan: Initial Draft
  - [ ] Spend Plan: Initial Draft
- [ ] Architecture Description Document (ADD)
  - [ ] Final
    - [ ] Introduction
    - [ ] Architecture Selection Criteria
    - [ ] Architecture Version
    - [ ] Architecture Standards and Regulations
    - [ ] Model Standards and Regulations
    - [ ] Supported Software and Operating Systems
- [ ] User & Analyst Manual (UAM)
  - [ ] Final
    - [ ] Introduction

- [ ] Model Functional Block Diagram
- [ ] Drafts
  - [ ] Threat System Description
  - [ ] Model Intended Use
  - [ ] Sources
  - [ ] Confidence Summary
  - [ ] Assumptions Limitations and Gaps
- [ ] Interface Control Document (ICD)
  - [ ] Draft
    - [ ] Model DLL Structs
- [ ] Model leads get concurrence from Tech Editor that documents are ready for SRR
- [ ] Model lead gets concurrence from Architecture Software Lead and Architecture Domain Lead that they're ready for SRR
- [ ] Model lead gets concurrence of TDL tech lead that the team is ready for SRR
- [ ] Model Lead signs off that the team is ready for SRR
- [ ] Determine required attendees for the meeting and provide to NGIC
  - [ ] Reserve Booz Allen conference room
- [ ] Briefing
- [ ] Programmatics Briefing
  - [ ] System Software Engineering Document: PM team will provide template
  - [ ] Review Quality Assurance Process: PM team will provide template

PDR: Virtual or In-person

- [ ] Architecture Description Document (ADD)
  - [ ] Drafts: Component Overviews, Component Code Structures
- [ ] User and Analyst Manual (UAM)
  - [ ] Final
    - [ ] Threat System Description
    - [ ] Model Intended Use
    - [ ] Sources
    - [ ] Confidence Summary
  - [ ] Drafts
    - [ ] Threat System CONOPS
    - [ ] Model Key Performance Indicators and Equations
    - [ ] Component Overview
    - [ ] Component Functional Behavior
    - [ ] Component Formulas and Equations
    - [ ] Component Key Performance Indicators
    - [ ] Component Parameters
    - [ ] Component Assumptions Limitations and Gaps
    - [ ] Model Parameters
    - [ ] Model Specific Log Files
    - [ ] Test Asset
- [ ] Test Description Document (TDD)
  - [ ] Drafts
    - [ ] Requirement Traceability Matrix
    - [ ] Tests Description

- [ ] Test Numbering Scheme
- [ ] Scope of Test Cases
- [ ] Test Family X
- [ ] Test Cases Shared Characteristics
- [ ] Test Family Default Inputs
- [ ] Test Case X1
- [ ] Behavior Tested
- [ ] Behavior Tested Laydown Diagram
- [ ] Prerequisites
- [ ] Test Inputs
- [ ] Test Procedure
- [ ] Expected Results & Evaluation Criteria

- [ ] Interface Control Document (ICD)
  - [ ] Draft
    - [ ] Model DLL Functions

- [ ] Requirements Spreadsheet: Final and Signed
- [ ] Programmatics completed by PM team
  - [ ] Project Plan: Update
  - [ ] Spend Plan: Update
- [ ] Antenna Pattern: Final
- [ ] Code Completion Items
  - [ ] ~20% Code Completion
  - [ ] ~100% component design outlined
- [ ] Model leads get concurrence from Tech Editor that documents are ready for PDR
- [ ] Model lead gets concurrence from Architecture Software Lead that they're ready for PDR
  - [ ] Implement/address all milestone code review feedback
- [ ] Model lead gets concurrence from Architecture Domain Lead that they're ready for PDR
  - [ ] Implement/address all milestone code review feedback
- [ ] Model lead gets concurrence of TDL tech lead that the team is ready for PDR
- [ ] Model Lead signs off that the team is ready for PDR
- [ ] Determine required attendees for the meeting and provide to NGIC
  - [ ] Reserve Booz Allen conference room
- [ ] Briefing
- [ ] Programmatics Briefing

CDR: Virtual or In-person

- [ ] Architecture Description Document (ADD)
  - [ ] Final
    - [ ] Component Overviews
    - [ ] Component Code Structures

- [ ] User and Analyst Manual (UAM)
  - [ ] Final
    - [ ] Assumptions Limitations and Gaps
    - [ ] Model Fidelity Log
    - [ ] Threat System CONOPS
    - [ ] Model Key Performance Indicators and Equations

- [ ] Component Overview
- [ ] Component Functional Behavior
- [ ] Component Formulas and Equations
- [ ] Component Key Performance Indicators
- [ ] Component Parameters
- [ ] Component Assumptions Limitations and Gaps
- [ ] Model Parameters
- [ ] Model Specific Log Files
- [ ] Test Asset

- [ ] Drafts
  - [ ] Model Specific Setup
  - [ ] Analytic Outputs

- [ ] Test Description Document (TDD)
  - [ ] Final
    - [ ] Requirement Traceability Matrix
    - [ ] Tests Description
    - [ ] Test Numbering Scheme
    - [ ] Scope of Test Cases
    - [ ] Test Family X
    - [ ] Test Cases Shared Characteristics
    - [ ] Test Family Default Inputs
    - [ ] Test Case X1
    - [ ] Behavior Tested
    - [ ] Behavior Tested Laydown Diagram
    - [ ] Prerequisites
    - [ ] Test Inputs
    - [ ] Test Procedure
    - [ ] Expected Results & Evaluation Criteria
  - [ ] Drafts
    - [ ] Assumptions and Constraints

- [ ] Interface Control Document (ICD)
  - [ ] Final
    - [ ] Model DLL Functions
    - [ ] Model DLL Structs

- [ ] Code Completion Items
  - [ ] ~80% Code Completion
  - [ ] ~100% Feature Implementation
  - [ ] Unit Testing Covering all major features, all pass
  - [ ] Some System tests, at least basic test and control test with results
- [ ] Antenna Pattern: Signed by NGIC
- [ ] Code Demo either at CDR or TRR (virtually via Teams or in-person) - discuss with TPOC
  - [ ] Demo dry-run with SERAPH Team Lead
- [ ] Model leads get concurrence from Tech Editor that documents are ready for CDR
- [ ] Model lead gets concurrence from Architecture Software Lead that they’re ready for PDR
  - [ ] Implement/address all milestone code review feedback
- [ ] Model lead gets concurrence from Architecture Domain Lead that they’re ready for PDR
  - [ ]

- [ ] Implement/address all milestone code review feedback

- [ ] Model lead gets concurrence of TDL tech lead that the team is ready for CDR
- [ ] Model Lead signs off that the team is ready for CDR
- [ ] Determine required attendees for the meeting and provide to NGIC
  - [ ] Reserve Booz Allen conference room
- [ ] Briefing
- [ ] Programmatics Briefing

TRR: Virtual

- [ ] User and Analyst Manual (TDD)
  - [ ] Final
    - [ ] Model Specific Setup
    - [ ] Analytic Outputs
- [ ] Test Description Document (TDD)
  - [ ] Final
    - [ ] Test Procedure
    - [ ] Expected Results & Evaluation Criteria
    - [ ] Unit Tests
  - [ ] Draft
    - [ ] Result Narratives
- [ ] Code Demo either at CDR or TRR (virtually via Teams or in-person) - discuss with TPOC
  - [ ] Demo dry-run with SERAPH Team Lead
- [ ] Model leads get concurrence from Tech Editor that documents are ready for TRR
- [ ] Model lead gets concurrence from Architecture Software Lead that they’re ready for PDR
  - [ ] Implement/address all milestone code review feedback
- [ ] Model lead gets concurrence from Architecture Domain Lead that they’re ready for PDR
  - [ ] Implement/address all milestone code review feedback
- [ ] Model lead gets concurrence of TDL tech lead that the team is ready for TRR
- [ ] Model Lead signs off that the team is ready for TRR
- [ ] Determine required attendees for the meeting and provide to NGIC
  - [ ] Reserve Booz Allen conference room
- [ ] Briefing
- [ ] Programmatics Briefing

FDR: In-person

- [ ] Government concurrence that all the above have been met/achieved
- [ ] Code
  - [ ] Source code
  - [ ] Release versions of the compiled code
  - [ ] Working GUI
- [ ] All test files, scripts, utilities, results, and methods, including log hash tests.
  - [ ] Unsigned copy of the requirements sheet showing the tests aligned to the requirements.
- [ ] ReadMe file containing below:
  - [ ] List of all documents and their file path locations
  - [ ] Quick start guide
  - [ ] Instructions to assemble and compile source code

- [ ] Final ICD
  - [ ] Final ICD with comments
- [ ] Final ADD
  - [ ] Final ADD with comments
- [ ] Final UAM
  - [ ] Final UAM with comments
- [ ] Final TDD
  - [ ] Final
    - [ ] Result Narratives
  - [ ] Final TDD with comments
- [ ] Model leads get concurrence from Tech Editor that documents are ready for FDR
- [ ] Current ODESSA UAM and ADD: ODESSA Architecture Documentation
- [ ] Copies of all intel used
- [ ] Final model classification and derivative classification authority block
- [ ] Copies of all previous milestone slides
- [ ] Reference ODESSA Delivery notes (WIP)
- [ ] Determine required attendees for the meeting and provide to NGIC
  - [ ] Reserve Booz Allen conference room
- [ ] Model demo
  - [ ] Demo dry-run with SERAPH Team Lead
- [ ] FDR briefing
- [ ] Updated programmatic plans have been provided to the appropriate government reps by the PM team